ATLAS NOTE

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Update of the Search for New Physics in the Dijet Mass Distribution in 163 pb^{-1} of pp Collisions at $\sqrt{s} = 7$ TeV Measured with the ATLAS Detector

The ATLAS Collaboration

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Abstract

We present a search for new physics in the dijet mass distribution using 163 pb^{-1} of data collected in 2011 with the ATLAS detector. In this update of the result described in [1] the analysis underwent only minor modifications, such as those due to changes in the trigger. No evidence for resonances was found in the updated sample. We improved the limits set on the product of cross section and acceptance for excited-quark (q^*) and axigluon production as a function of q^* and axigluon mass. These exclude at the 95% CL the q^* mass interval $0.80 < m_{q^*} < 2.49$ TeV and the axigluon mass interval $0.80 < m < 2.67$ TeV. Production cross section limits as a function of dijet mass are also set using a simplified Gaussian signal model to facilitate comparisons with other hypotheses.

1 Introduction

A search for new resonances and interactions manifested in two-jet final states was presented in [1], with 36 pb^{-1}of 7 TeV proton-proton collision data analyzed. We present an update of a portion of that analysis, the search using the dijet mass distribution, based on 163 pb^{-1} of data collected in 2011.

2 Changes from the previous version

Briefly summarizing the event selection, we require events where the two highest-p_{T} jets satisfy $|\eta_j| < 2.5$ with $|\Delta\eta_{jj}| < 1.3$. We also require that the two leading jets satisfy $p_{\mathrm{T}}^j > 30$ GeV, to guarantee high jet reconstruction efficiency, and impose a dijet mass cut to guarantee high trigger efficiency. The complete event selection criteria are given in [1] and have not been modified, except for the following:

- The lowest threshold unprescaled single jet trigger has changed, due to the increased LHC instantaneous luminosity. The nominal jet p_{T} threshold for the highest level of the trigger is 180 GeV.
- To attain full trigger efficiency, the dijet mass is required to be greater than 700 GeV (formerly 500 GeV).
- The quality criteria placed on the two leading jets to ensure that they arise from in-time energy deposition associated with proton-proton collisions have been revised slightly.

In 2011 the LHC has delivered increased instantaneous luminosity and, along with it, a significant number of in-time pile-up events (multiple pp collisions in the same bunch crossing) and out-of-time pile-up events (multiple pp collisions in nearby bunch crossings). The effects of this pile-up have been shown to be small for high-p_{T} jets such as those considered in this analysis. And, therefore, no special correction or uncertainty is applied.

3 Updated results

3.1 Background determination and search for dijet mass resonances

The background was extracted from the observed dijet mass spectrum by fitting the smooth function [2]

$$f(x) = p_1(1-x)^{p_2}x^{p_3+p_4\ln x}. \tag{1}$$

Figure 1 shows the updated mass spectrum and background. We also plot the significance of the bin-by-bin data-background difference: for bins where the discrepancy has a Poisson p-value of less than 0.5, the bin-by-bin significance of the data-background difference is shown, in units of σ (positive for excesses of data over background, negative for deficits).

Using a Poisson likelihood to compare the data and fitted background, and comparing the likelihood value to pseudo-experiments generated as statistical fluctuations of the fitted background distribution, we extract a p-value of 0.016. When we remove the four highest-mass events from the sample and re-fit, the p-value is 0.28. These four events were scanned by eye and showed no evidence of an origin as cosmic rays, beam halo, or detector anomalies.

Using the fitted background, we use the BumpHunter algorithm, which naturally incorporates the trials factor, to search for a resonance, as in [1]. The most significant bump is formed by the four bins from 3281 to 4230 GeV with a p-value of 0.25. Thus, there is no significant evidence of a resonance structure.

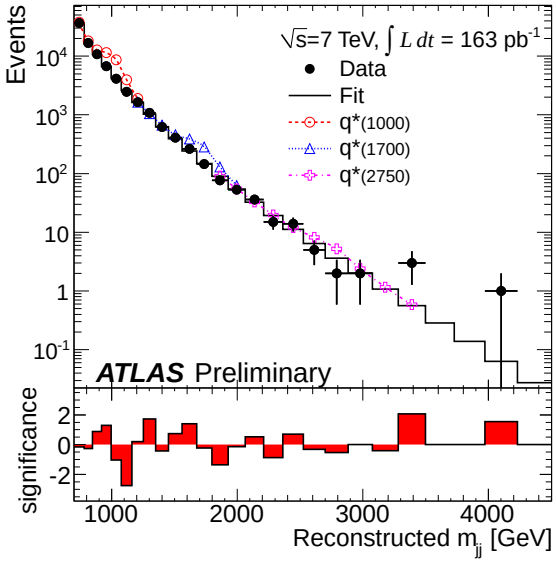

Figure 1: The data dijet mass distribution (filled points) fitted using a binned background distribution described by Eqn. 1 (histogram). The predicted q^* signals for excited-quark masses of 1000, 1700, and 2750 GeV are plotted on top of the background. For bins where the discrepancy has a Poisson p-value of less than 0.5, the bin-by-bin significance of the data-background difference is shown, in units of σ (positive for excesses of data over background, negative for deficits).

3.2 Limits on the production of new particles

In the absence of any observed significant discrepancy with the zero-signal hypothesis, the Bayesian method documented in [1] is used to set 95% credibility-level (CL) upper limits on $\sigma \cdot \mathscr{A}$, the product of the production cross section and acceptance for a hypothetical new particle decaying into dijets. The acceptance includes reconstruction and trigger efficiencies near 100%. The results are summarized in Fig. 2.

The observed and expected lower limits, with and without systematic uncertainty, at 95% CL, are summarized in Tables 1 and 2. We use the ATLAS default MC10 [3] Monte Carlo tune and MRST2007LO* [4] parton distribution functions.

We have included the effect of systematic uncertainties due to the luminosity, the background fit, and the jet energy scale. The effect of the uncertainty on the jet energy resolution was found to be negligible. The preliminary luminosity uncertainty for 2011 data is 4.5%. It is based on the 2010 luminosity calibration [5] which was transferred to the 2011 data by using the LAr forward calorimeter and the Tile calorimeter current measurements. The systematic uncertainty on the jet energy scale was estimated for 2010 data [6] and, in a preliminary fashion, this has been shown to still apply to 2011 data. The effect of in-time pile-up has been investigated and is considered to be negligible for jets with p_{T} greater than 100 GeV, such as those in this analysis. The effect of out-of-time pile-up has also been investigated, with no significant variation of dijet mass observed with respect to position in the LHC orbit, for the high mass events considered in this analysis. No additional uncertainty is included for the effects of pile-up. The background fit uncertainty is handled exactly as described in [1], by considering the range of background fits in pseudo-experiments.

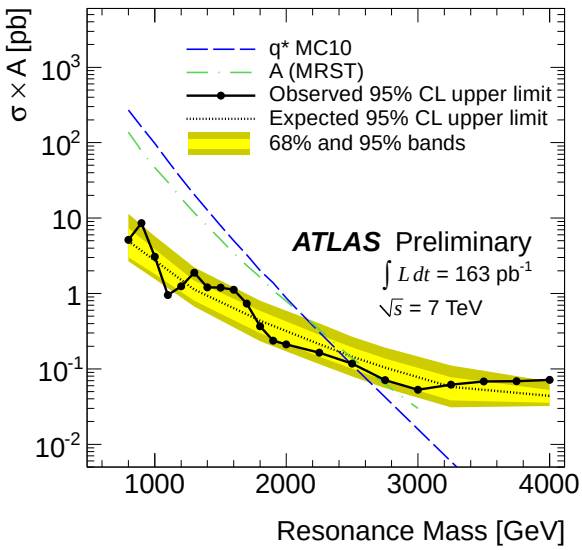

Figure 2: The 95% CL upper limit on $\sigma \cdot \mathscr{A}$ as a function of dijet resonance mass (black filled circles). The black dotted curve shows the expected 95% CL upper limit and the light and dark yellow shaded bands represent the 68% and 95% contours of the expected limit, respectively. The blue dashed and green dot-dashed curves represent $\sigma \cdot \mathscr{A}$ predictions for excited-quark and axigluon models, respectively.

Table 1: The 95% CL lower limits on the allowed q^* mass.

		Observed Mass Limit [TeV]		Expected Mass Limit [TeV]	
MC Tune	PDF Set	Stat. $\oplus$ Syst.	Stat. only	Stat. $\oplus$ Syst.	Stat. only
MC10 [3]	MRST2007LO* [4]	2.49	2.51	2.40	2.46

Table 2: The 95% CL lower limits on the allowed axigluon mass.

		Observed Mass Limit [TeV]		Expected Mass Limit [TeV]	
MC Tune	PDF Set	Stat. $\oplus$ Syst.	Stat. only	Stat. $\oplus$ Syst.	Stat. only
MC10 [3]	MRST2007LO* [4]	2.67	2.76	2.48	2.56

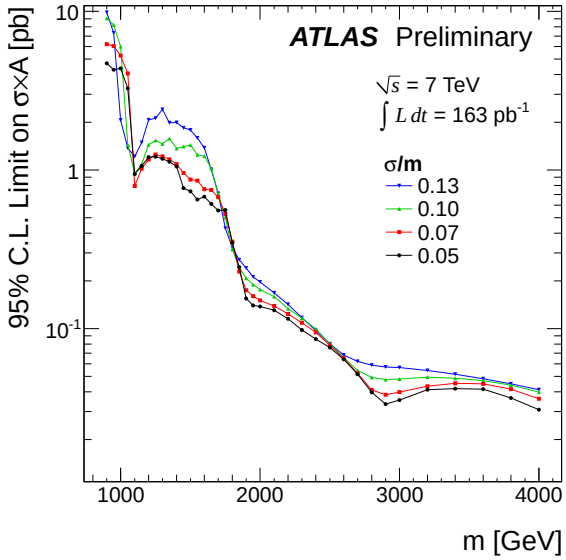

Figure 3: 95% CL limits on cross section times acceptance for hypothetical new particles with Gaussian reconstructed m_{jj} distributions, with systematic uncertainties included, for Gaussians of various widths, at each of various mean masses m. This is a graphical representation of Table 3.

3.3 Limits on simplified Gaussian models

As in [1] we also set limits on simplified Gaussian models, to facilitate comparisons with other hypotheses. We consider Gaussian distributions of reconstructed dijet mass, for various means and widths, and set 95% CL upper limits, including systematic uncertainties, on $\sigma \cdot \mathscr{A}$, the product of the production cross section and acceptance, for hypothetical models that would produce such distributions. These limits are presented in Table 3 and Fig. 3.

4 Summary

An update of the search for new particles decaying into dijets is presented, based on an integrated luminosity of 163 pb^{-1} collected with the ATLAS detector. No substantial changes were made to the analysis, except for the use of a different jet trigger, which resulted in a change in the minimum dijet mass to avoid trigger bias. No significant evidence of a resonance structure was found in this dataset. We exclude at the 95% CL the q^* mass interval $0.80 < m_{q^*} < 2.49$ TeV and the axigluon mass interval $0.80 < m < 2.67$ TeV. Production cross section limits as a function of dijet mass are also set using a simplified Gaussian signal model to facilitate comparisons with other hypotheses.

References

[1] ATLAS Collaboration, *Search for New Physics in Dijet Mass and Angular Distributions in pp Collisions at $\sqrt{s}$ = 7 TeV Measured with the ATLAS Detector*, New Journal of Physics **13** (2011) 053044, `arXiv:1103.3864 [hep-ex]`.

[2] CDF Collaboration, T. Aaltonen et al., *Search for new particles decaying into dijets in proton-antiproton collisions at $\sqrt{s} = 1.96$ TeV*, Phys. Rev. D **79** (2009) 112002.

Table 3: 95% CL limits on the cross section times acceptance (pb) for hypothetical new particles with Gaussian reconstructed m_{jj} distributions, with systematic uncertainties included. We present the widths as σ/m. A graphical version is shown in Fig. 3.

	σ/m			
Mean m (GeV)	0.05	0.07	0.10	0.13
900	4.7	6.2	9.1	9.9
1000	4.4	5.3	6.0	2.1
1100	0.95	0.79	0.95	1.2
1200	1.2	1.2	1.4	2.1
1300	1.2	1.2	1.5	2.4
1400	1.1	1.1	1.4	2.0
1500	0.73	0.87	1.4	1.8
1600	0.68	0.76	1.2	1.4
1700	0.55	0.67	0.71	0.71
1800	0.35	0.35	0.32	0.33
1900	0.15	0.17	0.21	0.24
2000	0.14	0.15	0.18	0.20
2100	0.13	0.14	0.16	0.17
2200	0.12	0.12	0.13	0.14
2300	0.098	0.11	0.12	0.12
2400	0.086	0.095	0.100	0.098
2500	0.076	0.078	0.080	0.080
2600	0.064	0.065	0.066	0.068
2700	0.052	0.052	0.055	0.062
2800	0.040	0.041	0.049	0.059
2900	0.033	0.038	0.048	0.057
3000	0.036	0.040	0.048	0.057
3200	0.041	0.043	0.049	0.055
3400	0.042	0.045	0.049	0.052
3600	0.042	0.045	0.047	0.048
3800	0.037	0.042	0.044	0.045
4000	0.031	0.036	0.040	0.041

[3] ATLAS Collaboration, *Charged particle multiplicities in pp interactions at $\sqrt{s}$ = 0.9 and 7 TeV in a diffractive limited phase space measured with the ATLAS detector at the LHC and a new PYTHIA6 tune*, ATLAS-CONF-2010-031 (unpublished, 2010) .

[4] A. Sherstnev and R. S. Thorne, *Parton Distributions for LO Generators*, Eur. Phys. J. C **55** (2008) 553–575, `arXiv:0711.2473 [hep-ph]`.

[5] ATLAS Collaboration, *Updated Luminosity Determination in pp Collisions at $\sqrt{s} = 7$ TeV using the ATLAS Detector*, ATLAS-CONF-2011-011 (unpublished, 2011) .

[6] ATLAS Collaboration, *Jet energy scale and its systematic uncertainty in proton-proton collisions at $\sqrt{s} = 7$ TeV in ATLAS 2010 data*, ATLAS-CONF-2011-007 (unpublished, 2011) .

Appendix: Additional Plots

Figure 5 is intended for possible use in a longer talk such as a seminar. Many such plots are publicly available on the ATLAS web pages along with the plots from [1]. Figure 5 is a new type of plot, in this spirit. The other Figures are, hopefully, self-explanatory.

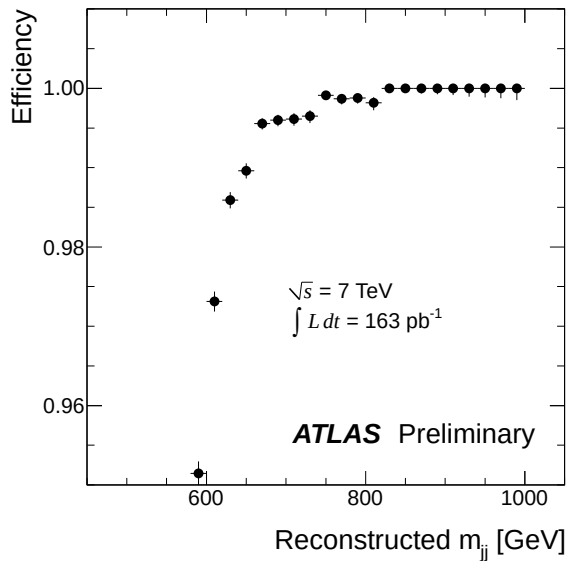

Figure 4: The efficiency for passing the primary trigger as a function of the dijet invariant mass. The uncertainties are statistical.

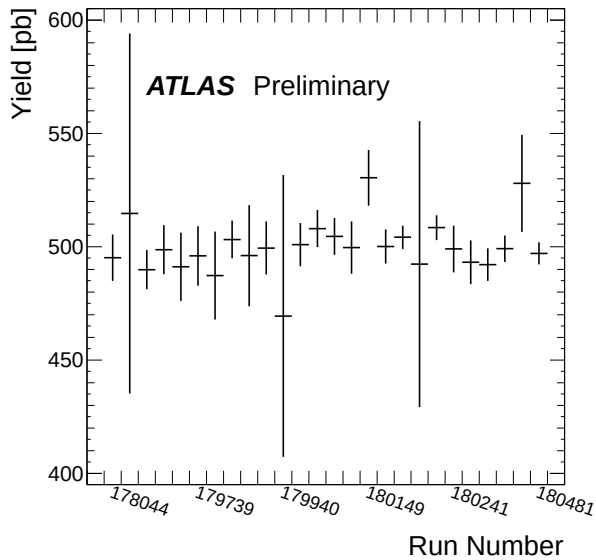

Figure 5: The yield, in pb, of events passing our event selection as a function of run number. The yield is consistent throughout the data-taking period.

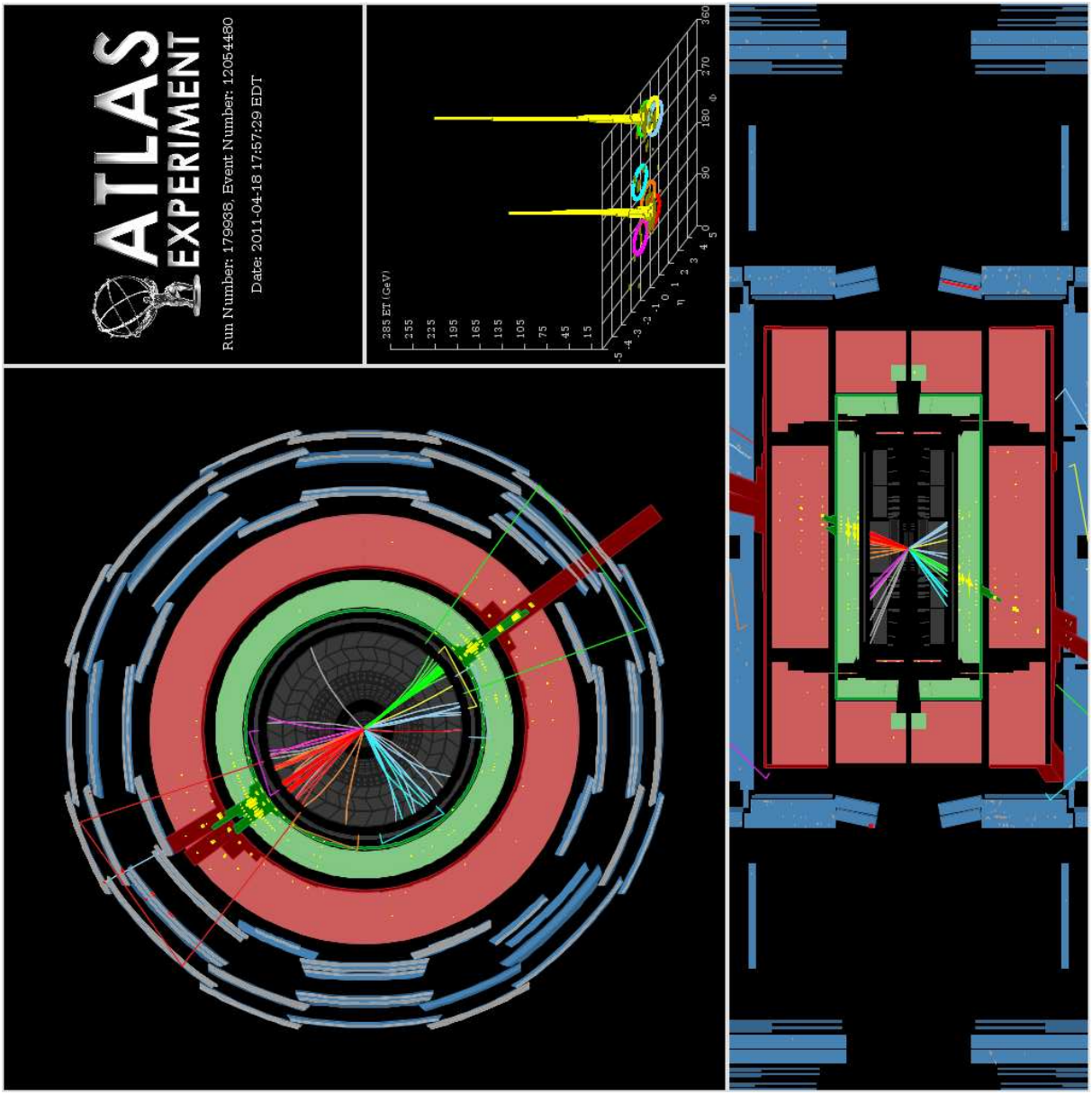

Figure 6: The highest mass event passing our event selection. The dijet mass is 4040 GeV while the leading jets have (p_{T} , η, ϕ, color) of (1850 GeV, 0.32, 2.2, red) and (1840 GeV, -0.53, -0.92, green).

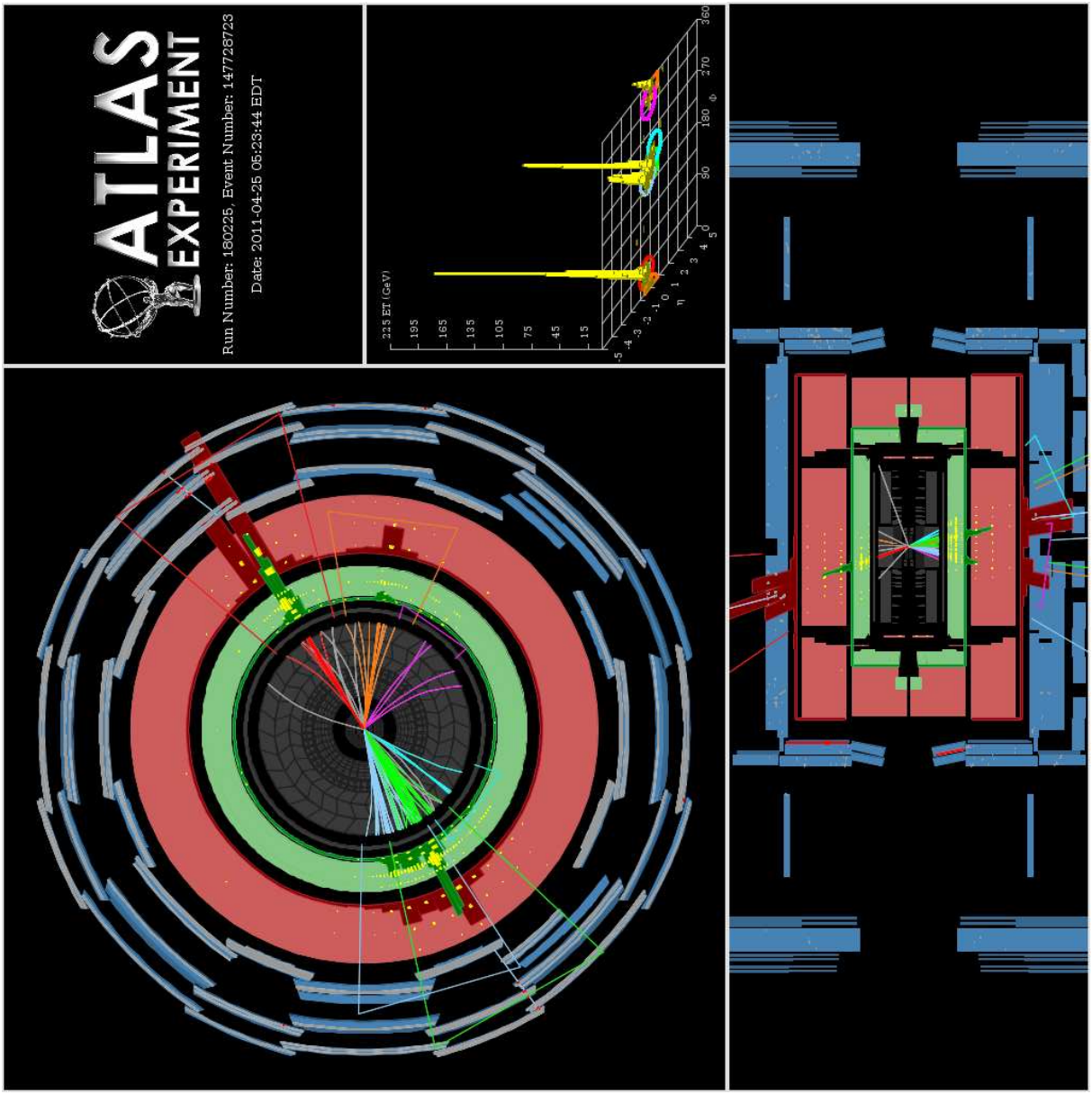

Figure 7: A high mass dijet event, with dijet mass 3360 GeV and leading jets with (p_{T} , η, ϕ, color) of (1640 GeV, 0.04, -2.7, green) and (1600 GeV, -0.35, 0.56, red).

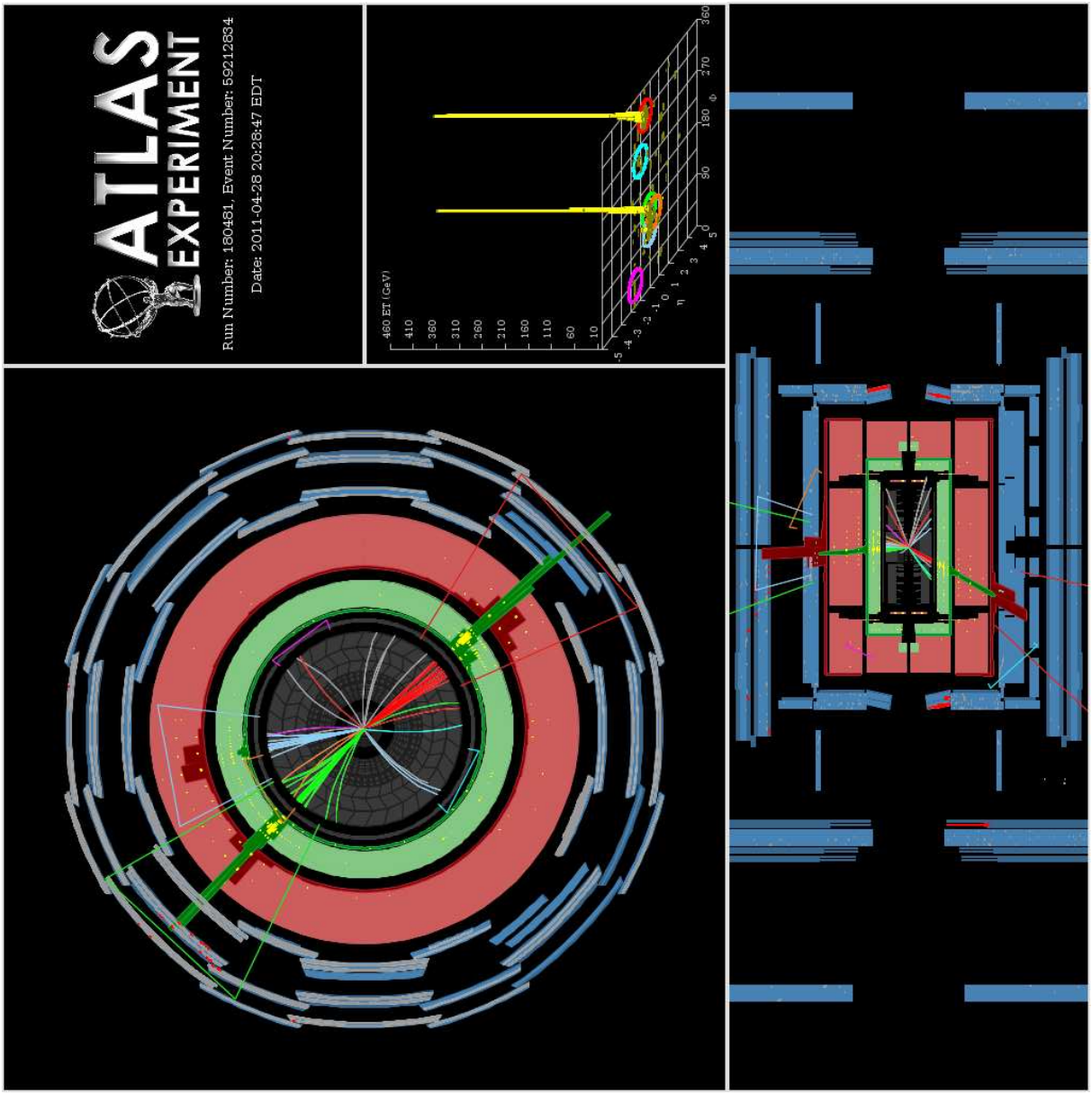

Figure 8: A high mass dijet event, with dijet mass 3350 GeV and leading jets with (p_{T} , η, ϕ, color) of (1800 GeV, -0.51, -0.85, red) and (1470 GeV, -0.05, 2.4, green).

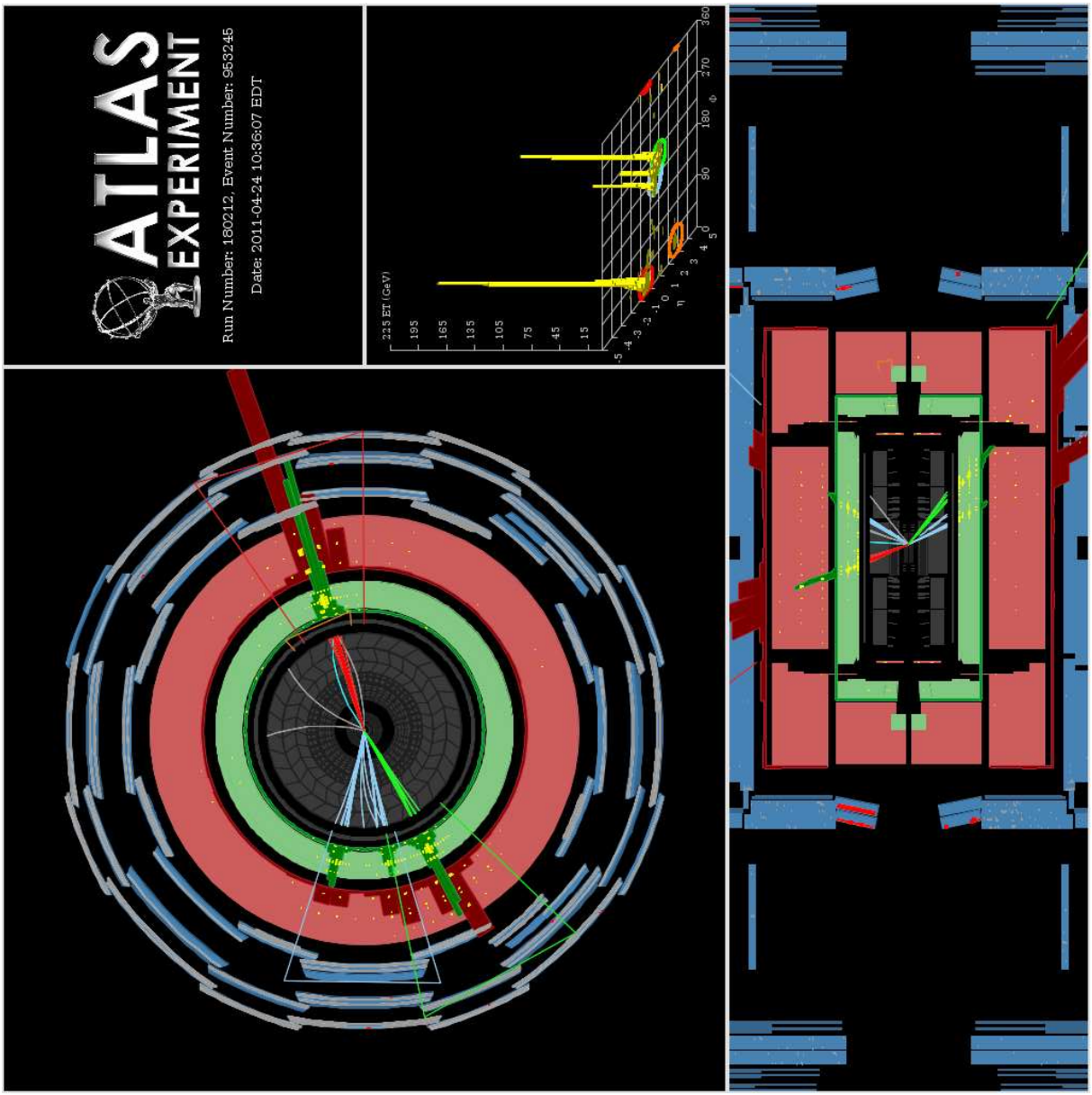

Figure 9: A high mass dijet event, with dijet mass 3320 GeV and leading jets with (p_{T} , η, ϕ, color) of (1590 GeV, -0.39, 0.31, red) and (1180 GeV, 0.88, -2.70, green).